Neural Machine Translation on Europarl English–French: A Controlled Comparison of LSTM+Attention and Transformer

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Abstract

We compare an LSTM with concat-style attention against a Transformer on English–French translation, using the Europarl corpus. For the matched 200K comparison, both models share the same SentencePiece-BPE tokenizer, train/val/test splits, and evaluation protocol. With 200K training pairs the Transformer reaches 30.43 ± 0.23 **test sacreBLEU** averaged over three seeds, against 6.59 **test BLEU** for the LSTM under the same preprocessing – a $+23.8$ BLEU gap. Pushing the Transformer to 500K pairs at the same architecture buys another $+3.95$ BLEU (test 34.56). The architectural gap is stable across seeds and grows on longer source sentences. Code, five trained checkpoints, and the per-run prediction CSVs are released.[1]

1 Introduction

Neural machine translation (NMT) has moved through several big steps in the last decade: the recurrent encoder–decoder of Sutskever et al. [2014], the attention mechanism of Bahdanau et al. [2015], and the Transformer of Vaswani et al. [2017]. The interesting question is not whether a Transformer beats an LSTM; published results already say it does. The interesting question is by how much, how stable that gap is across seeds, and how it compares to simply training the same architecture on more data. We answer those questions empirically.

Task and motivation. We use Europarl EN→FR [Koehn, 2005], the aligned record of European Parliament proceedings. It is a convenient testbed for this kind of comparison: alignment is clean, the register is formal but consistent, and a few hundred thousand pairs already produce non-trivial BLEU on a single Colab GPU.

Hypothesis. We state an explicit, testable claim:

> *H1.* At a fixed mid-resource budget on Europarl EN→FR, the architecture step from LSTM+Attention to Transformer produces a larger improvement in test BLEU than a $2.5\times$ increase in parallel training data at fixed Transformer architecture.

Contributions. The report contributes a fair LSTM-vs-Transformer comparison on Europarl EN→FR with shared SentencePiece-BPE tokenization [Kudo and Richardson, 2018], identical splits, and matched compute; a seed-variance estimate for the Transformer at 200K pairs (n=3,

[1] Code repository: CSE676 - Group Project - NMT ENFR.

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$\sigma = 0.23$ BLEU) that rules out seed luck as the explanation for the architectural gap; a $2.5\times$ data-scale ablation at fixed architecture; a length-stratified and qualitative error analysis; and a release of code, five trained checkpoints, and all per-run prediction CSVs so the experiments are reproducible.

2 Background and Related Work

The recurrent encoder–decoder of Sutskever et al. [2014] established the basic NMT recipe. Attention mechanisms [Bahdanau et al., 2015] made it work on long sentences; our baseline uses a concat-style variant of attention, described in §3.2. The Transformer [Vaswani et al., 2017] dropped recurrence for multi-head self-attention, which makes training parallel across positions and shortens the path between any two tokens to $O(1)$.

Byte-pair encoding [Sennrich et al., 2016] and its language-agnostic implementation SentencePiece [Kudo and Richardson, 2018] sidestep the open-vocabulary problem of earlier word-level NMT. We train SentencePiece on the training split only (§3). For evaluation we report sacreBLEU [Post, 2018, Papineni et al., 2002] on the official test split so the numbers are reproducible without tokenization ambiguity. Label smoothing [Szegedy et al., 2016] and Adam [Kingma and Ba, 2015] with linear warmup are the other standard ingredients we hold fixed across runs.

3 Method

3.1 Data and preprocessing

We use the Europarl EN–FR release. Each sentence pair is lowercased and kept only if both sides have between 1 and 50 whitespace tokens and the EN/FR length ratio sits in $[0.5, 2.0]$ – a simple filter that catches the worst alignment errors. We consider two corpus sizes, **200K** and **500K** raw source lines. For the 200K setting, filtering produced 180,289 sentence pairs, from which 174,289 were used for training and 3,000 each for validation and testing. The 500K setting yields 444,967 training pairs under the same filter, with 3,000-pair val and test holdouts drawn the same way. All models share the same splits.

Tokenization. A 16K-piece SentencePiece BPE model is trained on joint EN+FR text from the train split only; training it on val/test would leak test vocabulary. Special tokens <pad>, <unk>, <bos>, <eos> are ids 0–3. Maximum sequence length is 64 BPE tokens including BOS/EOS.

3.2 Models

LSTM+Attention (baseline). A bidirectional LSTM encoder (embedding dim 128, hidden dim 256, 2 layers, dropout 0.3) feeds a 2-layer LSTM decoder with a single-projection concat-style attention mechanism, where the decoder hidden state and encoder output are concatenated and passed through a learned linear projection followed by softmax. Forward and backward encoder states are concatenated to initialise the decoder. Total parameters: **26.3M**.

Transformer. A standard encoder–decoder Transformer [Vaswani et al., 2017] with d_{model}=512 and A single learned embedding matrix is tied between the encoder input, decoder input, and output projection. The Transformer and LSTM share the same SentencePiece BPE vocabulary, so both models see identical token streams. Sinusoidal positional encodings are added at the embedding layer. Total parameters: **52.3M**.

3.3 Training

Cross-entropy with pad ignored; the Transformer adds label smoothing [Szegedy et al., 2016] on the target distribution. The LSTM uses a scheduled-sampling curriculum [Bengio et al., 2015] that decays the teacher-forcing ratio linearly from 0.9 to a floor of 0.3. This is what causes its training loss to *rise* mid-run while validation loss keeps falling (§4.4). The LSTM uses Adam [Kingma and Ba, 2015]; the Transformer uses AdamW with a Noam schedule (warmup 4000 steps, peak learning rate $\approx 7 \times 10^{-4}$, followed by inverse-square-root decay). Both clip gradients at 1.0. The Transformer runs in bfloat16 mixed precision, putting a 200K-pair epoch at $\sim$85 s on a single Colab T4/A100-class GPU. Table 1 lists the rest of the hyperparameters.

Table 1: Training hyperparameters held fixed across runs.

	LSTM+Attention	**Transformer**
Tokenizer	SentencePiece BPE 16K	SentencePiece BPE 16K
Max sequence length	64 BPE tokens	64 BPE tokens
Batch size	64	128
Optimizer	Adam, lr 10^{-3}	AdamW, peak lr $\approx 7\times10^{-4}$
LR schedule	ReduceLROnPlateau	Noam: warmup 4000 steps, inverse-sqrt decay
Regularization	Dropout 0.3, weight decay 10^{-5}	Label smoothing 0.1
Teacher-forcing schedule	$\max(0.3,\ 0.9\ -\ 0.02\,(\text{epoch}\ -\ 1))$, floored at 0.3 from epoch 31 onward	N/A (parallel teacher forcing)
Precision	FP32	bfloat16 AMP
Gradient clip	1.0	1.0
Epochs / early-stop patience	35 / 7	15 / –
Parameters	26.3M	52.3M

Table 2: Main results. BLEU is sacreBLEU corpus BLEU; chrF is sacreBLEU character F-score on the same test split of 3,000 pairs (greedy decoding). "Time" is wall-clock per run on a single Colab-class GPU.

Run	**Arch**	**Train pairs**	**Seed**	**Params (M)**	**Val BLEU**	**Test BLEU**	**Test chrF**	**Time (min)**
LSTM 200K	LSTM+Attn	174,289	42	26.3	6.67	6.59	26.11	269
T 200K seed 42	Transformer	174,289	42	52.3	30.34	30.61	58.20	21
T 200K seed 13	Transformer	174,289	13	52.3	30.22	30.17	57.77	21
T 200K seed 99	Transformer	174,289	99	52.3	30.49	30.52	58.17	21
T 500K seed 42	Transformer	444,967	42	52.3	33.96	**34.56**	**61.00**	54
T 200K mean $\pm$ std	Transformer	174,289	–	52.3	30.35 ± 0.14	$\mathbf{30.43 \pm 0.23}$	58.05 ± 0.24	21

Decoding and evaluation. We evaluate with sacreBLEU [Post, 2018] corpus BLEU on detokenized hypotheses. The headline uses greedy decoding (max 80 BPE tokens). The headline uses greedy decoding with a maximum of 80 BPE tokens. The codebase also implements beam search with beam size 5 and length penalty 0.6, but we did not run a systematic beam-vs-greedy comparison, so greedy decoding is reported throughout for consistency.

4 Experiments

4.1 Setup

Five training runs (Table 2): three Transformer runs on 200K pairs with seeds 42, 13, 99 for the seed-variance estimate; one Transformer run on 500K pairs (seed 42) for the scale ablation; and an LSTM+Attention baseline on the same 200K pairs and the same tokenizer as the seed-42 Transformer, so the architectural comparison is matched.

4.2 Main results

The Transformer beats the LSTM by $+23.8$ test BLEU at matched data and preprocessing. Seed standard deviation is 0.23 BLEU, two orders of magnitude smaller than the gap, so the architectural effect is not seed noise.

4.3 Seed reliability and the data-scale ablation

The three 200K-pair Transformer runs land in $[30.17, 30.61]$ ($\sigma = 0.23$), so the 200K Transformer number is a tight anchor. Increasing the training budget by $2.5\times$ at the same architecture (Run *T 500K seed 42*) takes test BLEU to 34.56, a $+3.95$ BLEU bump. Side by side:

$$\Delta_{\text{architecture}}\ (\text{LSTM} \to \text{Transformer},\ 200\text{K})\ =\ +23.84\ \text{BLEU},\quad \Delta_{\text{data}}\ (200\text{K} \to 500\text{K},\ \text{Transformer})\ =\ +3.95\ \text{BLEU}.$$

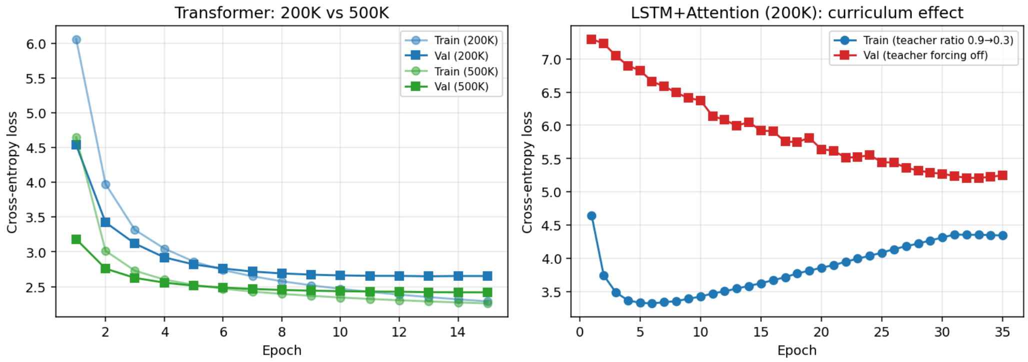

Figure 1: Train/validation loss curves. *Left:* Transformer at 200K vs 500K pairs – the 500K run reaches a markedly lower validation loss (final val 2.42 vs 2.65) and overfits less. *Right:* LSTM training loss rises after epoch $\sim$6 because the scheduled-sampling teacher-forcing ratio decays from 0.9 to 0.3, intentionally making training harder; validation loss, measured with teacher forcing off, keeps falling.

H1 (Section 1) holds: at this budget, picking the architecture matters roughly $6\times$ as much as adding $2.5\times$ more data.

4.4 Training dynamics

Figure 1 plots train and validation loss. The Transformer is the textbook shape – both curves drop monotonically, with mild overfitting after epoch $\sim$13 on 200K. The LSTM has a less common shape: *training loss rises after epoch* $\sim$6 *while validation loss keeps falling*. That is the curriculum at work. As the teacher-forcing ratio drops from 0.9 to 0.3, the model spends a larger share of each epoch decoding its own output instead of the ground-truth prefix, which is harder by construction. Validation loss is measured with teacher forcing off and is unaffected by the curriculum, so it tracks genuine generalization [Bengio et al., 2015].

4.5 Length-stratified analysis

To see *where* the Transformer wins, we bin test sentences by source length (English whitespace tokens) and compute sacreBLEU per bin (Figure 2). The shortest bucket is the interesting one. On 1–5 token source inputs, the LSTM scores 34.9 BLEU, roughly comparable to the Transformer-200K curve. The moment source length crosses 5 tokens it falls to 5–14 BLEU and stays there for the rest of the range. The Transformer's curve hardly moves; the 500K run sits in $[33.9, 35.8]$ across every bucket of ≥ 6 tokens. The likely reason is that self-attention provides $O(1)$ paths between any two positions, while the LSTM has to compress arbitrarily long context into a fixed-size recurrent state. The three Transformer-200K seeds overlap almost exactly per bucket, which lines up with the reliability finding above.

4.6 Qualitative comparison

Table 3 shows three test examples on which both models were evaluated. The cases shown are: a short input on which the LSTM enters a single-token repetition mode while the Transformer is fluent; a medium-length input where the LSTM's output is partially garbled (missing whitespace, fragmented morphemes) while the Transformer paraphrases the reference; and a longer input with a named entity, where the Transformer reproduces the entity and the surrounding clause faithfully while the LSTM collapses on the subword level. Bigram-level repetition is a recurring LSTM failure mode and is far less common in the 500K Transformer outputs. We do not report a quantitative repetition rate here.

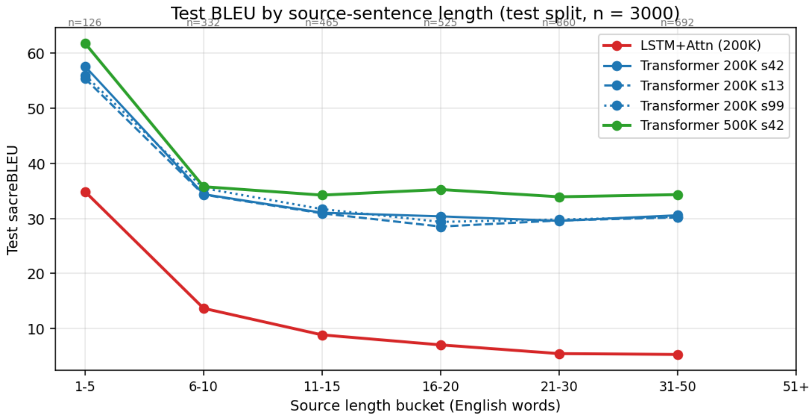

Figure 2: Test sacreBLEU stratified by source-sentence length (English whitespace tokens). The LSTM is competitive on ≤ 5-token inputs but collapses on longer sources; the Transformer's curve is roughly flat. Annotations above the panel are bucket sizes (test split, n=3000).

Table 3: Three qualitative test examples (greedy decoding). T-500K is the 500K-pair Transformer; LSTM is the 200K LSTM+Attention baseline. Because the LSTM and Transformer test splits were generated independently, qualitative examples are selected from the overlapping source sentences so that each row is a true same-source comparison. Sentence-BLEU is not reported here – the prediction CSVs only store `src`, `ref`, and `hyp`.

(a) Short input – LSTM single-token repetition.	
EN	that is why we should envisage imposing sanctions on israel.
REF	pour ces raisons, nous devrions envisager des sanctions à l'encontre d'israël.
T-500K	c'est pourquoi nous devrions envisager d'imposer des sanctions à israël.
LSTM	c'est pourquoi nous devrions sanctions sanctions sanctions sanctions sanctions.
(b) Medium input – LSTM garbled output; Transformer faithful.	
EN	perhaps we must also combat these things for the sake of our own future.
REF	peut-être devrons-nous aussi les combattre en vue de notre propre avenir.
T-500K	peut-être devons-nous également combattre ces choses pour notre propre avenir.
LSTM	nous devons égalementnous également de de nos propres-être-être.
(c) Long input with named entity – LSTM subword collapse.	
EN	we could invite mr tannock to ask this question during the next part-session, but if you wish to reply to it, you may do so.
REF	nous pourrions inviter m. tannock à formuler cette question pour la prochaine période de session, mais si vous désirez y répondre, vous pouvez le faire.
T-500K	nous pourrions inviter m. tannock à poser cette question lors de la prochaine période de session, mais si vous souhaitez y répondre, vous pouvez le faire.
LSTM	nous pouvons demander demander à m.annannannannannannannann cette question.

5 Discussion

The architecture step dominates the data step at this budget: $+23.8$ BLEU from LSTM to Transformer at 200K, versus $+3.95$ BLEU from $2.5\times$ more data at fixed Transformer architecture. Seed standard deviation is 0.23 BLEU, so the gap is not noise. The length-bucket result is consistent with the usual explanation – self-attention's $O(1)$ path length handles long context better than an LSTM's compressed recurrent state. We held tokenizer, splits, and decoding strategy fixed precisely so the architecture effect wouldn't be confounded by those.

What the $+3.95$ BLEU from doubling-and-then-some the data really tells us is harder to pin down from a single point. The LSTM is plausibly both data-starved *and* architecture-starved; the Transformer at 200K looks closer to architecture-saturated than data-saturated relative to its 500K counterpart. We can't say anything quantitative about diminishing returns beyond 500K from one data point – that would need 1M+ pairs and multi-GPU runs we didn't have the budget for.

6 Limitations and Broader Impact

Scope. Our claims cover EN→FR on Europarl with up to 500K parallel pairs, d_{model}=512, and greedy decoding. EN–FR is a high-resource pair and Europarl is a narrow, formal domain; neither the absolute BLEU nor the size of the architecture gap should be expected to transfer to a low-resource pair like EN→Swahili or to conversational test text.

Evaluation. We report sacreBLEU and chrF. BLEU correlates imperfectly with human judgment; a learned metric like COMET [Rei et al., 2020] would give a richer signal and is left as future work. chrF [Popović, 2015] mostly tracks BLEU here. The headline uses greedy decoding. The codebase also implements beam search with beam $= 5$ and length penalty $= 0.6$, but we did not run a systematic beam-vs-greedy comparison, so greedy decoding is reported throughout for consistency.

Bias and safety. Europarl's parliamentary register biases translations toward formal style and Eurocentric named entities. NMT models can hallucinate content not present in the source. The gendered French morphology against English's relatively gender-neutral pronouns is a known bias source – "the doctor said" tends to be translated with masculine agreement by default. The released system is a teaching artifact, not a production translation service.

Reproducibility. All five trained checkpoints, per-run prediction CSVs, training and analysis notebooks, and the SentencePiece tokenizers are released. Random seeds are recorded in each `metrics.json`. chrF is computed post-hoc using `sacrebleu.corpus_chrf` over each run's `test_predictions.csv` file.

7 Conclusion

On Europarl EN→FR under matched preprocessing, the Transformer beats LSTM+Attention by $+23.8$ BLEU (n=3 seeds, $\sigma = 0.23$). Going from 200K to 500K pairs at fixed Transformer architecture adds another $+3.95$ BLEU. Architecture matters more than data scale at this budget, and the gap is largest on long source sentences. We release the checkpoints, predictions, and notebooks so the comparison can be extended to other language pairs or richer metrics like COMET.

Use of Large Language Models

We used Claude (Anthropic) and ChatGPT (OpenAI) to understand the concepts behind NMT, LSTM and Transformers. We implemented the project with our own code with the understanding we gained from the LLMs.

Author Contributions

- **Matt Kattukuttiyil Das** – Data pipeline (Europarl loading, length-ratio filtering, splits), SentencePiece BPE tokenizer integration, Transformer training notebook, multi-seed experiments, analysis notebook, presentation slides, Q&A preparation length-bucket evaluation, report writing.
- **Souradeep Bhattacharya** – LSTM+Attention training notebook, scheduled-sampling curriculum, demo notebooks (Colab and local), checkpoint packaging and Drive hosting, Baseline (Checkpoint 1) notebook, Europarl Checkpoint 2 notebook, qualitative example curation.

All authors reviewed every notebook, contributed to experimental design, participated in the in-class presentation, and approved the final report.

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